AI-Powered SIEM Platform

UML Diagrams — Use Case & Activity Diagrams

Version 3.0 | April 2026

Security Information & Event Management

With LLM Threat Analysis, ML Anomaly Detection & Autonomous AI Agents

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Fig 1.1

Use Case / Procedure Diagram — Log Ingestion & Parsing

This use case diagram shows the interactions between external log sources (Winlogbeat, Syslog, Manual Ingestion), the Security Analyst, and the Log Ingestion subsystem. The system receives raw events through multiple interfaces, parses and normalizes them, stores in MongoDB, and provides query/search capabilities to the analyst.

Fig 1.2

Activity / Process Diagram — Log Processing Pipeline

This activity diagram shows the step-by-step flow from raw log reception through parsing, normalization, storage, and the initial handoff to the threat detection engine. It covers TCP/HTTP log receivers, the auto-parser with multi-format support, and MongoDB ingestion.

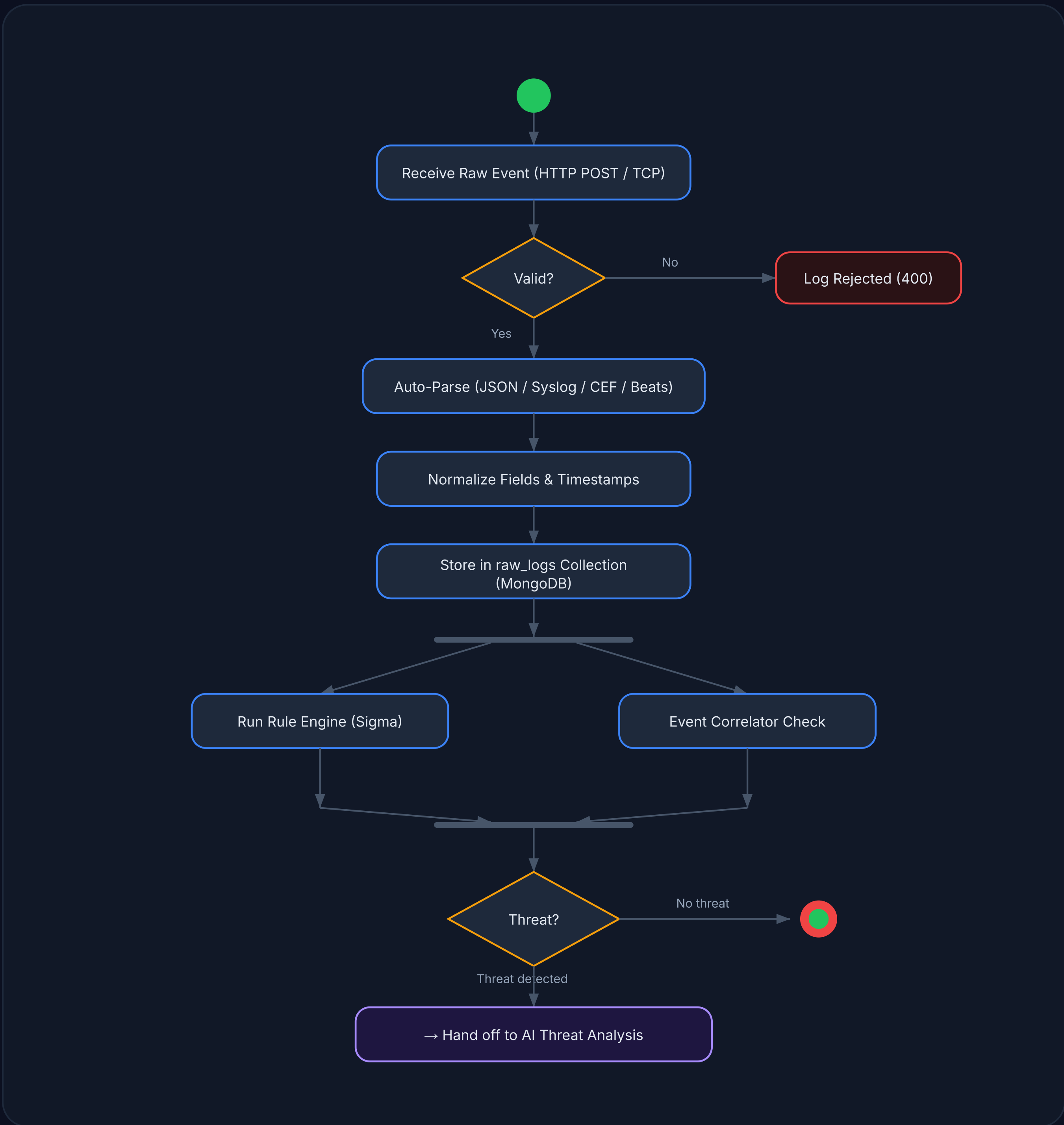

Fig 2.1

Use Case / Procedure Diagram — Threat Detection Engine

This diagram shows how the ML Model, LLM Engine (GPT-4/Gemini), Rule Engine, and Correlator interact with the Security Analyst to detect, classify, and enrich security threats. Includes MITRE ATT&CK mapping and GeoIP enrichment.

Fig 2.2

Activity / Process Diagram — AI Threat Analysis Pipeline

Complete activity flow from threat detection through ML scoring, LLM deep analysis, MITRE mapping, GeoIP enrichment to threat log creation and alert generation. Shows the multi-stage AI pipeline with conditional LLM invocation for high-severity events.

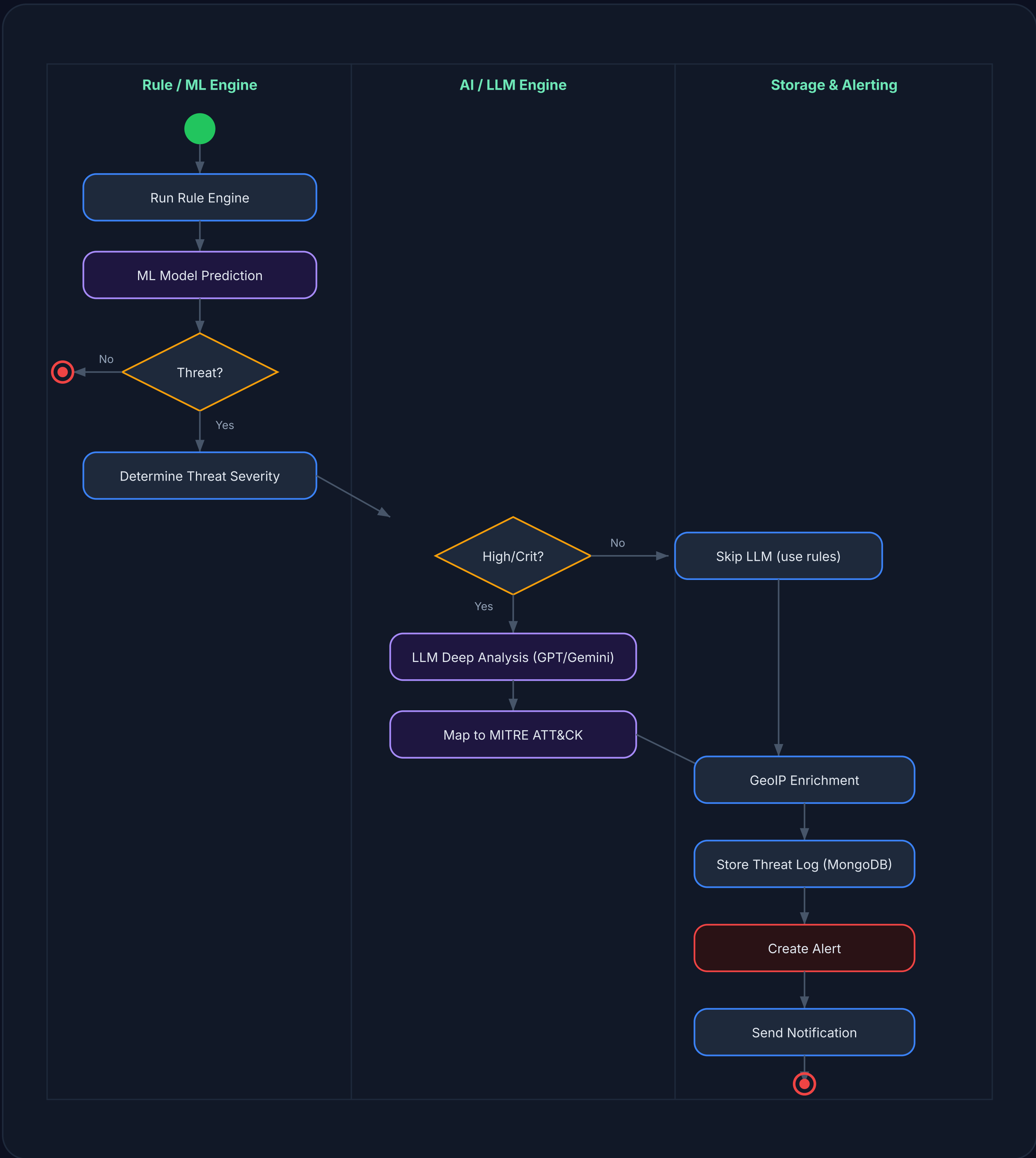

Fig 2.3

Use Case / Procedure Diagram — Alert Management System

This diagram covers the Alert Management lifecycle including viewing, filtering, investigating, updating status, and triggering AI agents from alerts. The SOC Analyst and the automated AI Agent system are the primary actors.

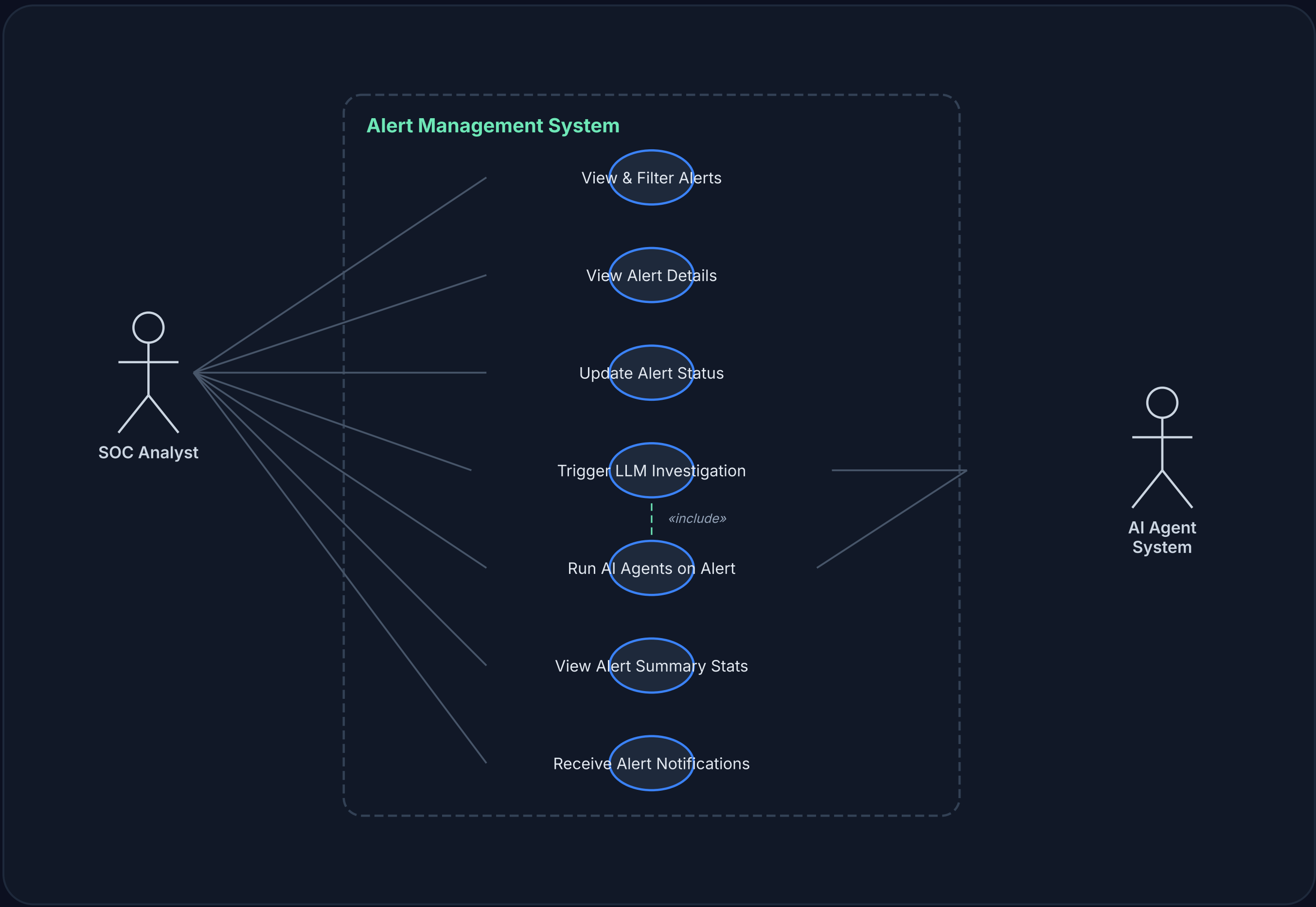

Fig 2.4

Activity / Process Diagram — Alert Lifecycle & Triage

This activity diagram traces the full alert lifecycle: from creation through triage, investigation, AI agent processing, response actions, to resolution. Shows the decision points where analyst and AI agents collaborate.

Fig 3.1

Use Case / Procedure Diagram — AI Agents & Orchestration

This diagram shows the AI Agent ecosystem: SOC Agent (triage), Analyst Agent (investigation), Responder Agent (mitigation), Forensics Agent (evidence collection), and Hunter Agent (proactive threat hunting). The Agent Manager orchestrates parallel execution.

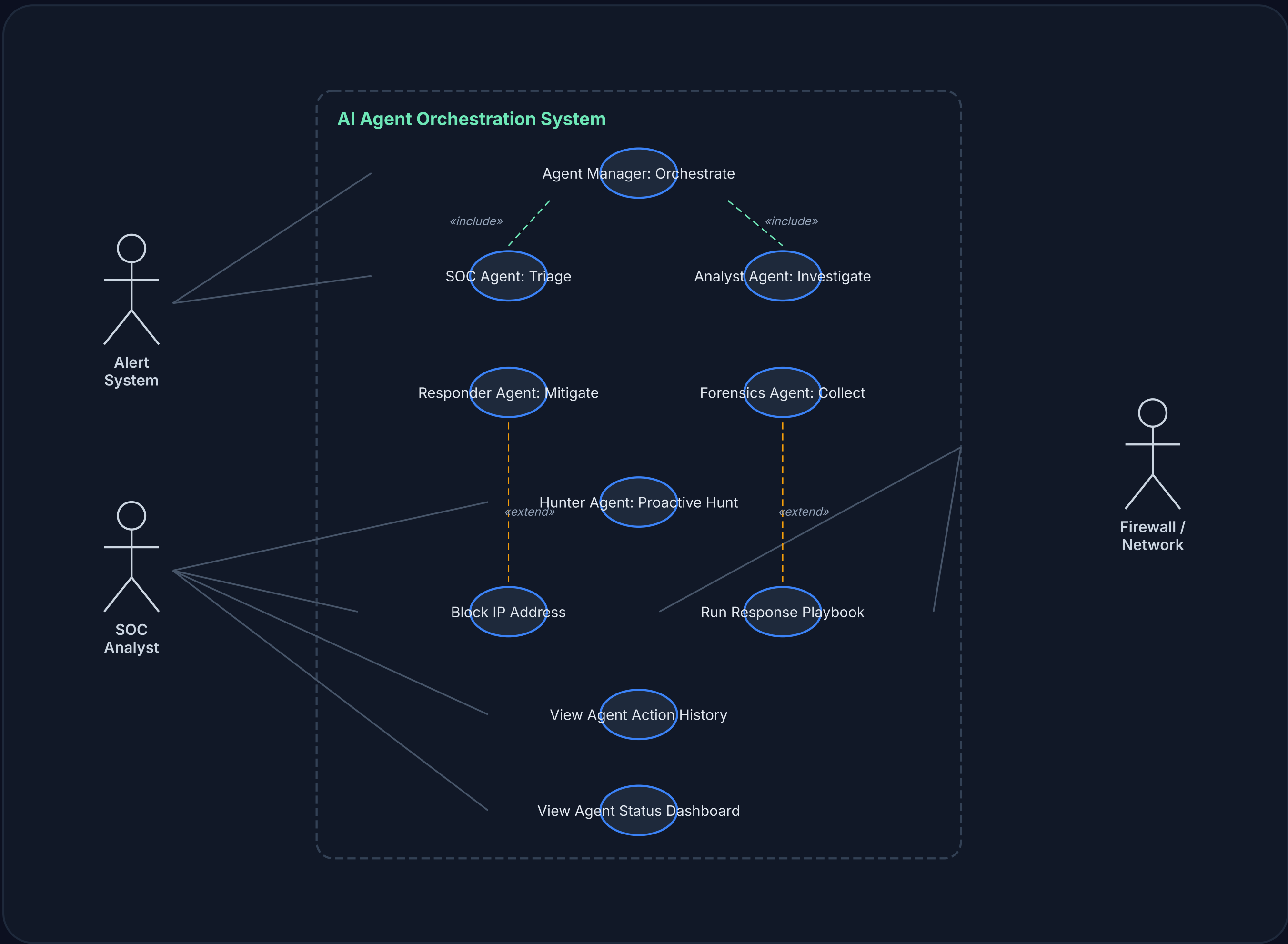

Fig 4.1

Activity / Process Diagram — Honeypot Deception System

This activity diagram shows the honeypot lifecycle: starting decoy services (SSH, HTTP, FTP, Telnet, SMB), capturing attacker connections, extracting credentials, analyzing payloads for risk indicators, and auto-generating alerts from honeypot hits.

Fig 4.2

Use Case / Procedure Diagram — Digital Forensics & Investigation

This diagram covers the Digital Forensics module including evidence collection, case management, timeline building, artifact analysis, reporting, and threat intelligence integration. The SOC Analyst and Forensic Investigator are the primary actors.

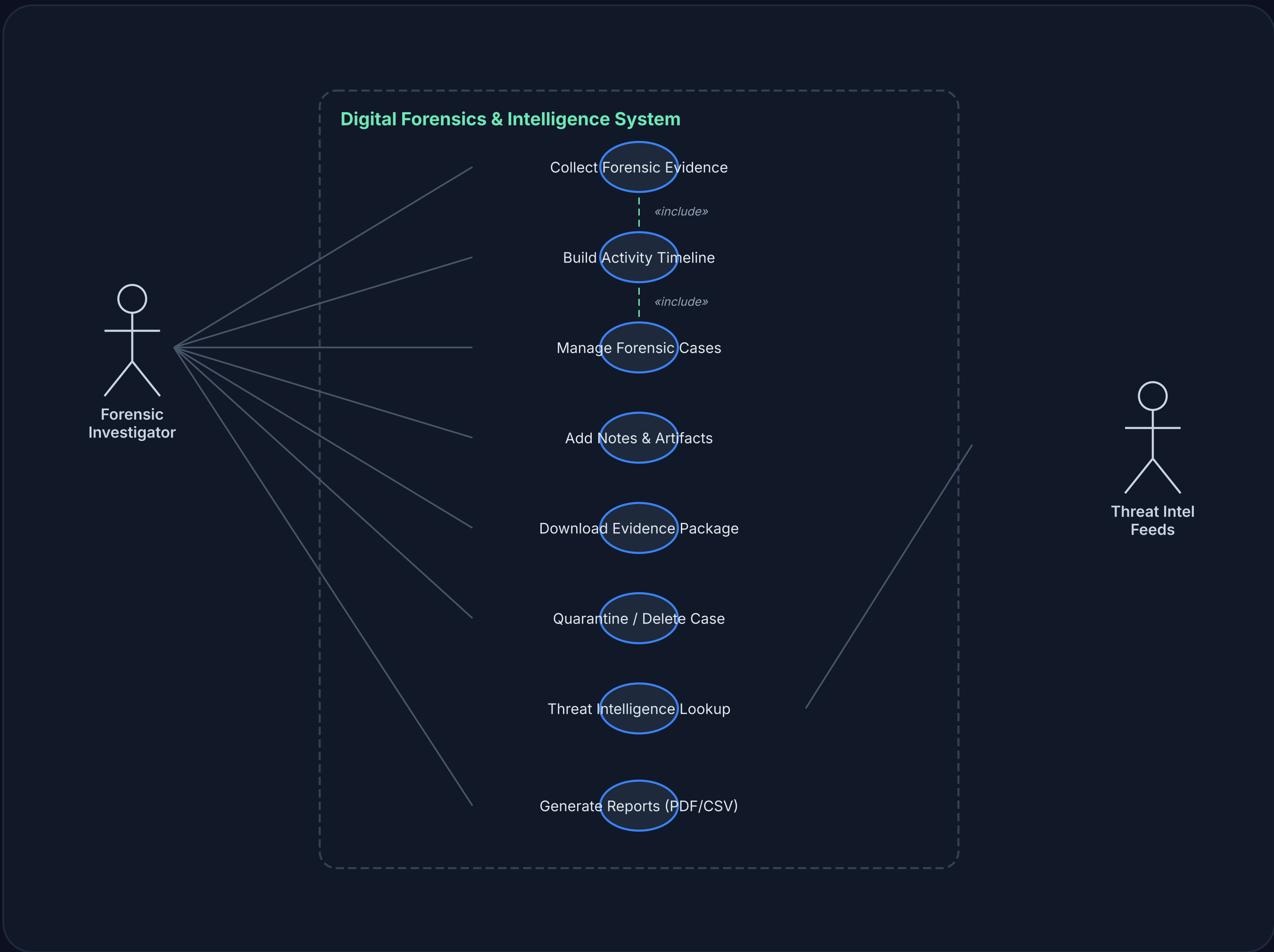

Fig 4.3

Activity / Process Diagram — Full System Integration Flow

This comprehensive activity diagram shows the end-to-end flow of the entire AI-Powered SIEM Platform: from log ingestion through threat detection, AI analysis, agent orchestration, automated response, forensic evidence collection, and final dashboard visualization. All major subsystems are represented.